

Let the Prototype Guide You: Robust Aggregation of Sparse Multi-Class Annotations via Annotator Prototype Learning

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Abstract

Truth inference is a critical technique for aggregating noisy and biased multi-class classification annotations. State-of-the-art approaches model each annotator using an individual confusion matrix. While well-grounded, they suffer from two fundamental bottlenecks: 1) confusion matrices are underfit when annotators label only a small subset of tasks or when classes are imbalanced, and 2) a single confusion matrix per annotator is inadequate for capturing complex annotator behaviors, leading to class-level collapse when tasks are extremely difficult. Simultaneously addressing these challenges is non-trivial, as it demands both robustness to data sparsity and sufficient expressiveness for complex annotator patterns. In this paper, we propose **CPBCC** (Class-specific Prototype-driven Bayesian Classifier Combination), which creatively models annotators through a dual-pathway architecture: (i) learning class-specific prototype annotation patterns across all annotators, and (ii) learning annotator-specific weights over prototypes. This framework addresses the bottlenecks and achieves a robust yet rich annotator characterization. Experiments across 10 real-world datasets spanning five domains demonstrate that CPBCC yields a 26% accuracy improvement in the best case, and boosts average accuracy from 68.73% to 74.11%. Our datasets and source code are available at https://github.com/JuJuCHEN-HHU/CPBCC_PTBCC.

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1. Introduction

Multi-class classification annotations are increasingly vital for advancing deep learning and fact-checking (Li, 2024; Otani et al., 2023; Chen et al., 2022a) in the era of large language models (Yao et al., 2024; Gienapp et al., 2025). However, such annotations are inherently noisy and biased (Hube et al., 2019; Li et al., 2019a) due to varying annotator reliability. To alleviate this, crowdsourcing platforms typically assign each task to multiple annotators (Sheng & Zhang, 2019; Wallace et al., 2022) and employ truth inference methods to derive the truth. The simplest approach, majority voting (MV), selects the most frequent annotation as the truth (Li et al., 2014a). While easy to implement, MV ignores annotator reliability variations, yielding sub-optimal results (Horn et al., 2018). Consequently, a wide range of sophisticated methods have been developed to infer annotator reliability and task truth iteratively.

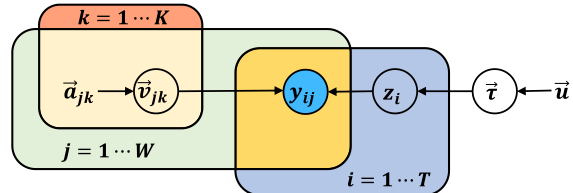


Figure 1. Plate of the representative confusion-matrix-based model IBCC (symbols are formally defined in Table 1).

Existing methods can be classified into three categories: single-parameter-based (Demartini et al., 2012; Li et al., 2019b; Cao et al., 2020), confusion-matrix-based (Dawid & Skene, 1979; Kim & Ghahramani, 2012; Li et al., 2019a; Zhang et al., 2024; Chen et al., 2025), and representation-learning-based approaches (Yin et al., 2017; Liu et al., 2024; 2025). Among these, confusion-matrix-based methods, which model each annotator with an individual confusion matrix, generally perform better and show great potential (Zhang et al., 2024). Figure 1 illustrates the plate of the IBCC model, a representative model of this category (Kim & Ghahramani, 2012). The DS model (Dawid & Skene, 1979) pioneered this approach. IBCC (Kim & Ghahramani, 2012) extend DS (Dawid & Skene, 1979) by incorporating Dirichlet priors. To capture annotator correlations, EBCC

(Li et al., 2019a) builds upon IBCC by clustering tasks with the same truth into subtypes. Moreover, Zhang et al. (2024) propose a coupled confusion correction method that trains two models simultaneously. Besides, Chen et al. (2025) replace the Dirichlet prior with a Softmax-Gaussian structure and update parameters via variational inference.

However, annotation sparsity and class imbalance are ubiquitous in practical scenarios (Ibrahim et al., 2023; Zhang et al., 2024; Xu et al., 2026), with multi-class settings posing substantially greater challenges than binary cases (Chu & Wang, 2021a; Yang et al., 2024; Zhang et al., 2025). Meanwhile, annotator behaviors are inherently more complex as well, which often systematically go beyond the representing capacity of a single confusion matrix. Therefore, while promising, *existing confusion-matrix-based methods suffer from two fundamental bottlenecks*: 1) the inferred confusion matrices are underfit when most annotators label only a small subset of tasks or when classes are imbalanced, and 2) a single confusion matrix is insufficient in capturing an annotator’s complex behaviors across all tasks, leading to class-level collapse when tasks are extremely difficult. Addressing these issues simultaneously is particularly challenging, as it demands both robustness to data sparsity and sufficient expressiveness for complex annotator patterns.

In response to the above issues, we propose **CPBCC** (Class-specific Prototype-driven Bayesian Classifier Combination). Specifically, CPBCC creatively captures annotator behaviors through a dual-pathway architecture: (i) learning class-specific prototype annotation patterns across all annotators, and ii) learning annotator-specific weights over these prototypes. Specifically, for each class, we assume that there exists a small set of prototypes, each capturing a canonical labeling pattern, and each annotator actually follows a personalized distribution over these prototypes. **The advantages are twofold.** First, by concentrating all data to train a small set of prototypes, we address the challenges of data sparsity and class imbalance. Second, representing each annotator as a personalized distribution over diverse prototypes allows for a richer annotator characterization. Extensive experiments on 10 widely used datasets demonstrate the clear superiority and scalability of our method.

The contributions of this study are summarized as follows:

- We identify two bottlenecks in state-of-the-art truth inference methods: underfitting from data sparsity or class imbalance, and limited expressiveness for complex annotator behaviors.
- We propose a novel class-specific prototype-driven framework that models annotators via a dual-pathway architecture: learning prototypes for each class and annotator-specific weights over prototypes.
- We develop a tractable approximation to the posterior.

- Extensive experiments across 10 real-world datasets spanning five domains demonstrate that we achieve a 26% accuracy improvement in the best case, and a 5.38% increase in average accuracy.

2. Related Work

Please refer to Appendix A for a detailed discussion of related work. Here we present a concise summary.

Depending on how annotators are modeled, existing works fall into single-parameter-based, confusion-matrix-based, and representation-learning-based approaches.

Single-parameter-based Methods assume that each annotator maintains a consistent reliability across tasks with different truths. Whitehill et al. (2009) model truth log-odds using a bilinear form of task difficulty and annotator reliability. Li et al. (2014b) model annotator errors using the Gaussian distribution and exploit the upper bound of the confidence interval to derive annotator reliability. Li et al. (2019b) and Cao et al. (2020) model annotator reliability using Gamma and scaled inverse chi-squared distribution, respectively, and updating them via the expectation-maximization and gradient descent method. Overall, most single-parameter-based methods are computationally efficient but less robust due to their strong homogeneity assumption.

Confusion-matrix-based Methods learn an individual confusion matrix for each annotator to capture class-specific reliability. This idea originates from the DS model (Dawid & Skene, 1979) and is extended by IBCC (Kim & Ghahramani, 2012) through the incorporation of Dirichlet priors. Subsequent works have enhanced IBCC by modeling annotator communities (Venanzi et al., 2014), task subtypes (Li et al., 2019a), matrix factorization and annotator co-occurrences (Ibrahim et al., 2019; Ibrahim & Fu, 2021), detecting annotator copying behaviors (Song et al., 2021), and exploring alternative parameterizations (Chen et al., 2025). While confusion-matrix-based methods generally perform better and show strong potential, they are sensitive to data sparsity and class imbalance and are limited in capturing the complex annotator patterns.

Representation-learning-based Methods leverage representation learning to encode relations among annotators, tasks, and annotations. Yin et al. (2017) propose LAA, which integrates a classifier-encoder and a reconstructor-decoder, and treats annotations as latent features. Wu et al. (2023a) introduce a multi-view graph embedding model over the task-annotator graph, and propose a heritable mechanism for annotator estimation. Liu et al. (2024) propose a graph contrastive framework in which data augmentation and a contrastive objective are introduced to encourage invariant representations across views. Generally, representation-learning-based methods excel at capturing

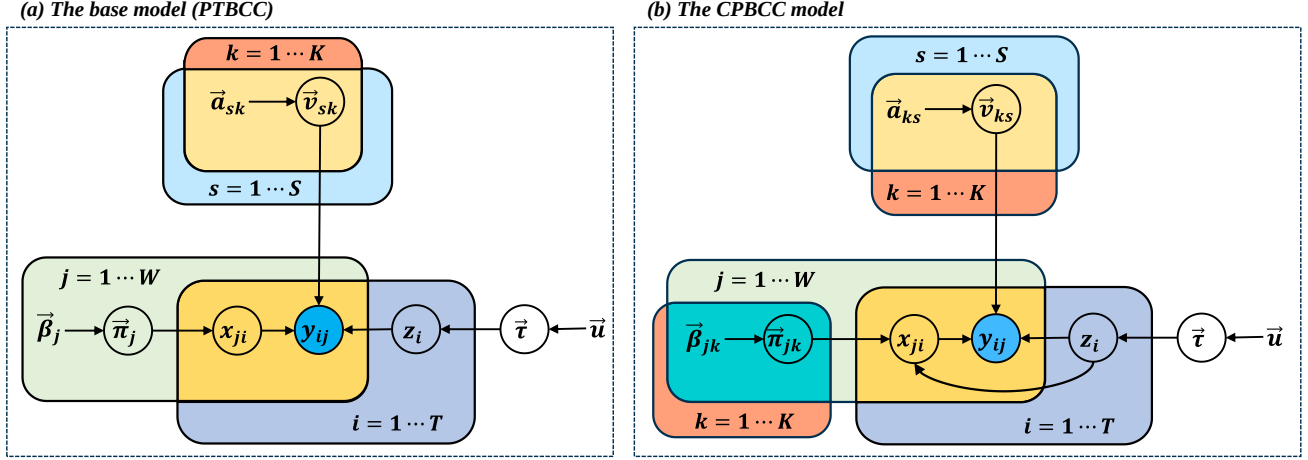


Figure 2. Plates of the proposed PTBCC and CPBCC model, respectively.

complex relationships (Liu et al., 2024; Li et al., 2025). However, their reliance on abundant data for network training limits their effectiveness in sparse settings.

3. Problem Definition

Basic mathematical notations are defined in Table 1.

We consider a crowdsourcing setting with T tasks, W annotators, and K classes. Let $y_{ij} \in \{1 \dots K\}$ denote the annotation provided by w_j ($j \in \{1 \dots W\}$) for t_i ($i \in \{1 \dots T\}$), and $z_i \in \{1 \dots K\}$ denote the truth of t_i . The objective of truth inference is to infer the latent truth z_i for each t_i . This involves iterative estimates of: (i) the truth probability of each annotation, and (ii) the reliability of each annotator.

Table 1. Basic mathematical notations.

Symbol	Description
T, W, K	Number of tasks, annotators, and classes
S	Number of prototypes
t_i	The i -th task
w_j	The j -th annotator
s	Index of prototype
k	Index of class
z_i	Truth of t_i
y_{ij}	Label provided by w_j for t_i
$\vec{v}_{jk}$	Annotator w_j 's label distribution given truth k
$\vec{v}_{sk}$	Prototype s 's label distribution given truth k
$\vec{v}_{ks}$	For truth k , prototype s 's label distribution
$\vec{a}_{jk}$	Dirichlet prior for $\vec{v}_{jk}$
$\vec{\tau}$	Truth distribution
$\vec{u}$	Dirichlet prior for $\vec{\tau}$
$\mathcal{W}_i$	Set of annotators who have annotated t_i
$\mathcal{T}_j$	Set of tasks that w_j have annotated

4. Method

4.1. Motivation

The representative confusion-matrix-based method IBCC (Figure 1) estimates a $K \times K$ confusion matrix for each w_j , with v_{jkl} ($k, l \in \{1 \dots K\}$) denoting the probability that w_j labels l while the truth is k . To address the bottlenecks caused by data sparsity, class imbalance, and limited expressiveness, we first propose the base **PTBCC** (ProtoType-driven Bayesian Classifier Combination) model, as illustrated in Figure 2(a). Instead of learning an individual confusion matrix per annotator, PTBCC assumes that annotators are accommodated by a small set of prototype confusion matrices, each capturing a canonical labeling pattern. Annotators are captured by their personalized distribution over these prototype confusion matrices, enabling shared statistical strength and individual heterogeneity.

The generative process of PTBCC is as follows:

1. Draw truth distribution:

$$\vec{\tau} | \vec{u} \sim \text{Dirichlet}(\vec{u})$$

2. For each prototype s ($s \in \{1 \dots S\}$):

- 2.1. Given class k ($k \in \{1 \dots K\}$) being the truth:

- Draw annotation distribution:

$$\vec{v}_{sk} | \vec{a}_{sk} \sim \text{Dirichlet}(\vec{a}_{sk})$$

3. For each annotator w_j ($j \in \{1 \dots W\}$):

- Draw prototype distribution:

$$\vec{\pi}_j | \vec{\beta}_j \sim \text{Dirichlet}(\vec{\beta}_j)$$

4. For each task t_i ($i \in \{1 \dots T\}$):

- 4.1. Draw truth assignment:

$$z_i | \vec{\tau} \sim \text{Categorical}(\vec{\tau})$$

- 4.2. For each annotator $w_j \in \mathcal{W}_i$:
 - 4.2.1. Draw prototype assignment:
 $x_{ji} | \vec{\pi}_j \sim \text{Categorical}(\vec{\pi}_j)$
 - 4.2.2. Draw annotation assignment:
 $y_{ij} | \vec{v}_{x_{ji}z_i} \sim \text{Categorical}(\vec{v}_{x_{ji}z_i})$

Please refer to Appendix B for inference details. While PTBCC enhances robustness in data-sparse and class-imbalanced settings, its *confusion-matrix-level* prototypes can be too restrictive, as annotator behaviors may vary substantially across classes. Motivated by this, we propose CPBCC to further overcome this shortcoming.

4.2. Class-specific Prototype-driven BCC

As illustrated in Figure 2(b), **CPBCC** (Class-specific Prototype-driven Bayesian Classifier Combination) relaxes PTBCC by modeling the more fine-grained *class-specific* prototypes. This development enables annotators more flexibility while retaining the advantages of prototype-based parameter sharing.

The generative process of CPBCC is as follows:

1. Draw truth distribution:
 $\vec{\tau} | \vec{u} \sim \text{Dirichlet}(\vec{u})$
2. For each class k ($k \in \{1 \dots K\}$) being the truth:
 - 2.1. For each prototype s ($s \in \{1 \dots S\}$):
 - Draw annotation distribution:
 $\vec{v}_{ks} | \vec{a}_{ks} \sim \text{Dirichlet}(\vec{a}_{ks})$
3. For each annotator w_j ($j \in \{1 \dots W\}$):
 - 3.1. For each class k ($k \in \{1 \dots K\}$) being the truth:
 - Draw prototype distribution:
 $\vec{\pi}_{jk} | \vec{\beta}_{jk} \sim \text{Dirichlet}(\vec{\beta}_{jk})$
4. For each task t_i ($i \in \{1 \dots T\}$):
 - 4.1. Draw truth assignment:
 $z_i | \vec{\tau} \sim \text{Categorical}(\vec{\tau})$
 - 4.2. For each annotator $w_j \in \mathcal{W}_i$:
 - 4.2.1. Draw prototype assignment:
 $x_{ji} | \vec{\pi}_j \sim \text{Categorical}(\vec{\pi}_j)$
 - 4.2.2. Draw annotation assignment:
 $y_{ij} | \vec{v}_{z_i x_{ji} z_i} \sim \text{Categorical}(\vec{v}_{z_i x_{ji} z_i})$

Following this generative process, the joint distribution is:

$$\begin{aligned}
 & p(Y, Z, V, X, \pi, \tau | u, \beta, a) \\
 &= p(\tau | u) p(\pi | \beta) p(V | a) p(Z | \tau) p(X | \pi, Z) p(Y | X, Z, V) \\
 &= \text{Dir}(\vec{\tau} | \vec{u}) \cdot \prod_j \prod_k \text{Dir}(\vec{\pi}_{jk} | \vec{\beta}_{jk}) \cdot \prod_k \prod_s \text{Dir}(\vec{v}_{ks} | \vec{a}_{ks}) \\
 &\cdot \prod_i \tau_{z_i} \prod_{j \in \mathcal{W}_i} \pi_{j z_i x_{ji} z_i} v_{z_i x_{ji} z_i y_{ij}}
 \end{aligned} \tag{1}$$

4.3. Inference with Variational Methods

Due to the intractability of exact inference, we apply mean-field variational inference (Bishop, 2006) to approximate the posterior. The key idea is to rewrite the log evidence as:

$$\begin{aligned}
 & \log p(C) \\
 &= \int q(B) \log \frac{p(C, B)}{q(B)} dB + \int q(B) \log \frac{q(B)}{p(B|C)} dB \\
 &= L(q) + KL(q(B) || p(B|C))
 \end{aligned} \tag{2}$$

where C and B denote the observed data and latent variables, respectively. Then, the posterior can be approximated by maximizing $L(q)$ under the mean-field strategy,

$$q(B) = \prod_{h=1}^H q_h(B_h) \tag{3}$$

and we can achieve the optimal $q_h^*(B_h)$:

$$q_h^*(B_h) \propto \exp(\mathbb{E}_{q(B_{-h})} \log p(C, B)) \tag{4}$$

Therefore, the goal is to find a variational distribution q ,

$$\begin{aligned}
 & q(Z, V, X, \pi, \tau) \\
 &= \text{Dir}(\vec{\tau} | \vec{u}) \cdot \prod_j \prod_k \text{Dir}(\vec{\pi}_{jk} | \vec{\eta}_{jk}) \cdot \prod_k \prod_s \text{Dir}(\vec{v}_{ks} | \vec{\mu}_{ks}) \\
 &\cdot \prod_i q(z_i | \vec{\phi}_i) \prod_{j \in \mathcal{W}_i} q(x_{ji} | \vec{\theta}_{ji z_i})
 \end{aligned} \tag{5}$$

To this end, the lower bound of the log evidence is:

$$\begin{aligned}
 & \log L(q) = \sum_k (u_k + \sum_i \phi_{ik} - \nu_k) \mathbb{E}_q \log \tau_k + \log B(\vec{v}) \\
 &+ \sum_j \sum_k \left\{ \sum_s (\beta_{jks} + \sum_{i \in \mathcal{T}_j} \phi_{ik} \theta_{jiks} - \eta_{jks}) \mathbb{E}_q \log \pi_{jks} \right. \\
 &+ \log B(\vec{\eta}_{jk}) \left. \right\} + \sum_k \sum_s \left\{ \sum_l (a_{ksl} - \mu_{ksl} \right. \\
 &+ \sum_j \sum_{i \in \mathcal{T}_j} \phi_{ik} \theta_{jiks} \mathbf{1}[y_{ij} = l]) \mathbb{E}_q \log v_{ksl} + \log B(\vec{\mu}_{ks}) \left. \right\} \\
 &- \sum_i \sum_k \phi_{ik} \log \phi_{ik} - \sum_j \sum_{i \in \mathcal{T}_j} \sum_k \phi_{ik} \sum_s \theta_{jiks} \log \theta_{jiks} \\
 &+ \text{Const}
 \end{aligned} \tag{6}$$

where $\mathbf{1}[\cdot]$ is an indicator function that returns 1 if $[\cdot]$ is true and 0 otherwise.

4.4. Parameter Update

We sequentially derive each term in Equation (5) by maximizing Equation (6). Following the standard mean-field

update steps, the update rules are:

$$\nu_k = u_k + \sum_i \phi_{ik} \quad (7)$$

$$\eta_{jks} = \beta_{jks} + \sum_{i \in \mathcal{T}_j} \phi_{ik} \theta_{jiks} \quad (8)$$

$$\mu_{ksl} = a_{ksl} + \sum_j \sum_{i \in \mathcal{T}_j} \phi_{ik} \theta_{jiks} \mathbf{1}[y_{ij} = l] \quad (9)$$

$$\theta_{jiks} \propto \exp(\mathbb{E}_q \log \pi_{jks} + \mathbb{E}_q \log v_{ksy_{ij}}) \quad (10)$$

$$\begin{aligned} \phi_{ik} &\propto \exp(\mathbb{E}_q \log \tau_k + \sum_{j \in \mathcal{W}_i} \sum_s \theta_{jiks} (\mathbb{E}_q \log \pi_{jks} \\ &+ \mathbb{E}_q \log v_{ksy_{ij}})) \end{aligned} \quad (11)$$

and the expectations of the log parameters are:

$$\mathbb{E}_q \log \tau_k = \psi(\nu_k) - \psi\left(\sum_k \nu_k\right) \quad (12)$$

$$\mathbb{E}_q \log \pi_{jks} = \psi(\eta_{jks}) - \psi\left(\sum_s \eta_{jks}\right) \quad (13)$$

$$\mathbb{E}_q \log v_{ksl} = \psi(\mu_{ksl}) - \psi\left(\sum_l \mu_{ksl}\right) \quad (14)$$

where $\psi(\cdot)$ denotes digamma function. We update all parameters iteratively until the change in each ϕ_{ik} falls below a threshold ξ .

4.5. Theoretical Analysis

Let $\mathcal{L}(\hat{Z}, Z) = \frac{\sum_i (\hat{Z} \neq Z)}{T}$ denote the average model error,

Theorem 4.1. *The minimax error rates of PTBCC and CPBCC are lower bounded by Eq. (15) and Eq. (16), respectively*

$$\begin{aligned} \inf_{\hat{Z}} \sup \mathbb{E}[\mathcal{L}(\hat{Z}, Z)] &\geq -\frac{\log 2}{T^2 \log K} + \frac{1}{T^2 \log K} \sum_i [H(\vec{\phi}_i) \\ &- \sum_{j \in \mathcal{W}_i} \sum_{k, k'} \phi_{ik} \phi_{ik'} \sum_s \pi_{js} \text{KL}(\vec{v}_{sk} \| \vec{v}_{sk'})] \end{aligned} \quad (15)$$

$$\begin{aligned} \inf_{\hat{Z}} \sup \mathbb{E}[\mathcal{L}(\hat{Z}, Z)] &\geq -\frac{\log 2}{T^2 \log K} + \frac{1}{T^2 \log K} \sum_i [H(\vec{\phi}_i) \\ &- \sum_{j \in \mathcal{W}_i} \sum_{k, k'} \phi_{ik} \phi_{ik'} \sum_s \pi_{jks} \text{KL}(\vec{v}_{ks} \| \vec{v}_{k's})] \end{aligned} \quad (16)$$

where $H(\vec{\phi}_i) = \sum_k \phi_{ik} \log \phi_{ik}$.

Please refer to Appendix C for the proof. Theorem 4.1 provides a principled characterization of the key to truth inference through the class-level KL divergence between induced annotation distributions. Modeling individual confusion matrix per annotator is inadequate in capturing complex annotator behaviors and blurs inter-class distinctions, reducing KL divergence and yielding a looser lower bound. In

contrast, our dual-pathway architecture maintains larger KL distances between classes, tightening the lower bound and systematically alleviating the bottlenecks of state-of-the-art methods. Specifically, CPBCC strictly generalizes PTBCC and standard confusion-matrix models: PTBCC is recovered by tying prototype parameters across classes, and a standard confusion-matrix model is recovered by collapsing the prototype mixture. More importantly, when annotator behavior is genuinely class-dependent, as is common in real-world scenarios and verified by the experiments, class-agnostic parameterizations are misspecified, whereas CPBCC can exactly cover the scenarios.

5. Experiments

5.1. Experimental Setup

Datasets. We use 10 widely adopted real-world datasets spanning five domains for evaluation. Among them, CF, MS, Dog, Face, Adult, and Web are also used in Li et al. (2019b); Val5, Val7, and Aircr are also used in Wu et al. (2023b); and LabelMe also appears in Liu et al. (2024). Table 2 summarizes their statistics.

Metrics. We evaluate the results of accuracy and Macro-F1.

Initialisation. We set $\xi = 0.001$, initialize $\{\phi_{ik}\}$ by majority voting, and set $S = 2$ for PTBCC, $S = 3$ for CPBCC.

For PTBCC, we set $(f, m) = (5, 1.35)$, $v_{1kl} = 1$, $v_{1kk} = f$, $v_{2kl} = m$, $v_{2kk} = 1$, and then row normalize $\vec{v}_{sk}$. $\{\theta_{jis}\}$, $\{\beta_{jks}\}$, and $\{a_{ksl}\}$ are set via variational rules.

Notably, PTBCC is evaluated under a unified initialization across all datasets and serves as our foundational model. By examining the two prototypes it learns from the data, we can gain a clear and intuitive understanding of the overall annotator behaviors.

By analyzing the results learned by PTBCC, we observed that the two prototype matrices for datasets Val5 and Val7 are not clearly diagonal-dominant. Therefore, for CPBCC, we do not impose a strong prior structure when initializing the parameters for Val5 and Val7; instead, we allow the model to learn from the data automatically. Specifically, we set $b = 5$, $v_{ksl} = 1$, and $v_{ksk} = b$, then row-normalize $\vec{v}_{ks}$, and set $\vec{\pi}_{jk} = [0.5, 0.2, 0.01]$; $\{\theta_{jiks}\}$, $\{\beta_{jks}\}$, and $\{a_{ksl}\}$ are determined via variational rules. For other datasets, we set $d = 0.35$, $\vec{\theta}_{jik} = [\phi_{ik}, d, 0.10]$, and then normalize $\vec{\theta}$; $\{\beta_{jks}\}$ are determined using the last answered task, and $\{a_{ksl}\}$ are determined via variational rules.

Baselines. We include MV, DS (Dawid & Skene, 1979), GLAD (Whitehill et al., 2009), IBCC (Kim & Ghahramani, 2012), CBCC (Venanzi et al., 2014), CATD (Li et al., 2014b), BWA (Li et al., 2019b), EBCC (Li et al., 2019a), TiReMGE (Wu et al., 2023a), and GOVERN (Liu et al.,

Table 2. Statistics of 10 real-world datasets; $\text{Min}(w_j)$ and $\text{Median}(w_j)$ are the minimum and median number of tasks labeled by annotators.

Dataset	T	W	K	$\text{Min}(w_j)$	$\text{Median}(w_j)$	#Y	Truth distribution	Domain
Val7	100	38	7	20	20	1,000	{0.08,0.21,0.18,0.10,0.23,0.19,0.01}	Sentiment
Aircr	593	50	6	1	4	1,588	{0.17,0.17,0.13,0.17,0.17,0.19}	Image
CF	300	461	5	1	2	1,720	{0.19,0.23,0.24,0.31,0.03}	Sentiment
MS	700	44	10	2	16	2,945	{0.11,0.10,0.10,0.09,0.10,0.10,0.10,0.11,0.11,0.09}	Audio
Dog	807	109	4	1	38	8,070	{0.21,0.23,0.27,0.29}	Image
Face	584	27	4	4	56	5,242	{0.25,0.25,0.25,0.25}	Image
Adult	11,040	825	4	1	18	89,799	{0.56,0.18,0.11,0.15}	Level
Val5	100	38	5	20	20	1,000	{0.13,0.27,0.23,0.28,0.09}	Sentiment
Web	2,665	177	5	1	19	15,567	{0.12,0.18,0.26,0.20,0.23}	Relevance
LabelMe	1,000	59	8	3	27	2,547	{0.09,0.12,0.13,0.11,0.14,0.13,0.13,0.15}	Image

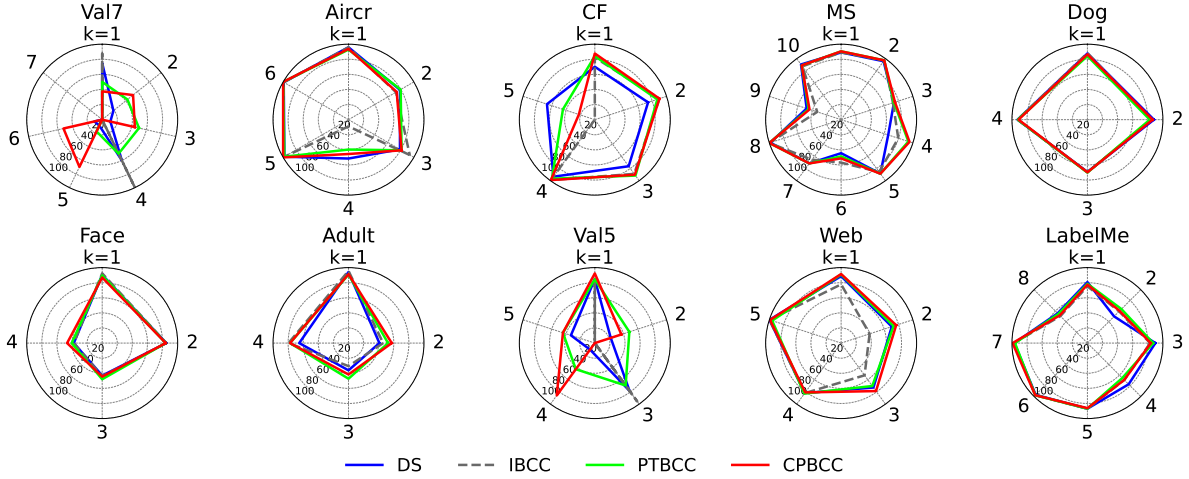


Figure 3. Class-wise accuracy results of CPBCC, PTBCC, and representative confusion-matrix-based methods, DS and IBCC.

2024). More details are given in Appendix A.

We repeat all experiments 10 times on a server with 2 vCPUs (Intel Xeon Platinum 8369HC) and 8 GB of memory.

5.2. Effectiveness

Table 3 presents performance results across different datasets and baselines. Figure 3 illustrates class-wise accuracy results of CPBCC, PTBCC, and representative confusion-matrix-based methods, DS and IBCC. As shown in Table 3, our CPBCC and PTBCC consistently achieve top-tier accuracy, with CPBCC holding an advantage over PTBCC.

Handling Data Sparsity and Class Imbalance. Val7 and Val5 exhibit both class imbalance and extreme task difficulty, where current methods struggle significantly with an accuracy no more than 22% and 41%, respectively. By modeling annotators through class-specific prototypes and annotator-specific weights over prototypes, our CPBCC and PTBCC boost the accuracy to 48%, 31% for Val7, and 50%, 56% for Val5, respectively. DS, as the simplest confusion-

matrix-based method, exhibits stronger generalization but ranks last on the sparse ($\text{Median}(w_j)=2$) and imbalanced CF dataset, indicating the restrictive requirements for both robustness to data sparsity and sufficient expressiveness for complex annotator behaviors. Meanwhile, GOVERN and TiReMGE achieve competitive performance, highlighting the potential of representation learning. However, they are also sensitive to data sparsity and class imbalance, i.e., GOVERN performs worse on Adult, while TiReMGE performs worse on Web.

Sufficient Expressiveness. Notably, even in data-abundant and class-balanced settings such as Face, our methods remain competitive. As shown in Figure 3, on class 4, CPBCC achieves 46.5% accuracy and PTBCC 40.4% versus 37.6% for both IBCC and DS, suggesting our structural superiority in annotator characterization. Moreover, on Web dataset, CPBCC achieves 85.11% overall accuracy versus DS’s 82.55% and IBCC’s 70.41%, with dramatic per-class advantages: CPBCC wins on 3 out of 5 classes (91.5%, 76.7%, 78.6% on classes 1, 2, and 3 versus DS’s 88.7%, 70.6%, 72.8%, while IBCC catastrophically degrades, collapsing to 39.7% on class 2 and 52.8% on class 3.

Table 3. Performance results under different datasets and baselines.

Accuracy (mean% $\pm$ std.%) $\uparrow$											
Method	Val7	Aircr	CF	MS	Dog	Face	Adult	Val5	Web	LabelMe	Avg.
MV	19.90 $\pm$ 0.94	81.01 $\pm$ 0.65	88.73 $\pm$ 0.53	70.17 $\pm$ 0.65	82.33 $\pm$ 0.29	63.61 $\pm$ 0.49	75.86 $\pm$ 0.49	35.30 $\pm$ 2.83	73.19 $\pm$ 0.57	76.55 $\pm$ 0.40	66.67 $\pm$ 0.33
CATD	16.00 $\pm$ 0.00	80.29 $\pm$ 0.46	87.47 $\pm$ 0.86	78.69 $\pm$ 0.59	82.11 $\pm$ 0.22	61.66 $\pm$ 0.05	75.08 $\pm$ 0.23	33.00 $\pm$ 0.00	80.32 $\pm$ 0.05	76.78 $\pm$ 0.36	67.14 $\pm$ 0.17
DS	22.00 $\pm$ 0.00	84.15$\pm$0.00	79.67 $\pm$ 0.00	76.43 $\pm$ 0.00	84.26$\pm$0.00	64.04 $\pm$ 0.00	73.87 $\pm$ 0.00	41.00 $\pm$ 0.00	82.55 $\pm$ 0.00	79.30$\pm$0.00	68.73 $\pm$ 0.00
GLAD	20.00 $\pm$ 0.00	80.61 $\pm$ 0.00	88.00 $\pm$ 0.00	78.43 $\pm$ 0.00	83.52 $\pm$ 0.00	63.01 $\pm$ 0.00	76.28 $\pm$ 0.00	33.00 $\pm$ 0.00	79.87 $\pm$ 0.00	77.00 $\pm$ 0.00	67.97 $\pm$ 0.00
IBCC	17.00 $\pm$ 0.00	78.08 $\pm$ 0.00	88.33 $\pm$ 0.00	76.43 $\pm$ 0.00	83.77 $\pm$ 0.00	64.73 $\pm$ 0.00	<u>77.18$\pm$0.00</u>	33.00 $\pm$ 0.00	70.41 $\pm$ 0.00	77.50 $\pm$ 0.00	66.64 $\pm$ 0.00
CBCC	13.00 $\pm$ 0.00	76.56 $\pm$ 0.00	89.33 $\pm$ 0.00	77.29 $\pm$ 0.00	80.79 $\pm$ 0.00	65.41 $\pm$ 0.00	75.08 $\pm$ 0.00	33.00 $\pm$ 0.00	70.67 $\pm$ 0.00	76.50 $\pm$ 0.00	65.76 $\pm$ 0.00
BWA	20.00 $\pm$ 0.00	82.46 $\pm$ 0.00	89.33 $\pm$ 0.00	78.57 $\pm$ 0.00	83.15 $\pm$ 0.00	61.82 $\pm$ 0.00	74.17 $\pm$ 0.00	35.00 $\pm$ 0.00	82.25 $\pm$ 0.00	77.40 $\pm$ 0.00	68.42 $\pm$ 0.00
TiReMGE	22.00 $\pm$ 0.45	80.25 $\pm$ 0.19	88.90 $\pm$ 0.26	72.09 $\pm$ 0.16	82.49 $\pm$ 0.32	64.32 $\pm$ 0.24	76.58 $\pm$ 0.00	37.40 $\pm$ 0.49	67.03 $\pm$ 0.71	77.15 $\pm$ 0.17	66.82 $\pm$ 0.09
EBCC	16.70 $\pm$ 0.78	81.32 $\pm$ 0.07	88.33 $\pm$ 0.00	78.71 $\pm$ 0.00	<u>84.01$\pm$0.00</u>	63.46 $\pm$ 0.78	74.77 $\pm$ 0.13	33.40 $\pm$ 0.49	74.37 $\pm$ 0.00	78.60 $\pm$ 0.00	67.37 $\pm$ 0.15
GOVERN	14.70 $\pm$ 2.19	81.85 $\pm$ 0.46	<u>89.67$\pm$0.84</u>	77.83 $\pm$ 0.50	82.29 $\pm$ 0.90	66.56$\pm$0.88	69.52 $\pm$ 1.49	32.10 $\pm$ 2.77	89.66$\pm$1.70	77.56 $\pm$ 0.83	68.17 $\pm$ 0.53
PTBCC	31.00 $\pm$ 0.00	82.29 $\pm$ 0.00	89.33 $\pm$ 0.00	<u>79.43$\pm$0.00</u>	82.40 $\pm$ 0.00	<u>65.58$\pm$0.00</u>	<u>77.18$\pm$0.00</u>	56.00$\pm$0.00	81.94 $\pm$ 0.00	78.80 $\pm$ 0.00	72.40 $\pm$ 0.00
CPBCC	48.00$\pm$0.00	<u>82.80$\pm$0.00</u>	90.33$\pm$0.00	79.86$\pm$0.00	83.40 $\pm$ 0.00	<u>65.58$\pm$0.00</u>	77.78$\pm$0.00	<u>50.00$\pm$0.00</u>	<u>85.11$\pm$0.00</u>	78.20 $\pm$ 0.00	74.11$\pm$0.00
Macro-F1 (mean% $\pm$ std.%) $\uparrow$											
Method	Val7	Aircr	CF	MS	Dog	Face	Adult	Val5	Web	LabelMe	Avg.
MV	17.80 $\pm$ 0.83	79.73 $\pm$ 0.71	79.11 $\pm$ 1.60	70.37 $\pm$ 0.30	82.13 $\pm$ 0.37	61.88 $\pm$ 0.60	63.02 $\pm$ 0.83	31.61 $\pm$ 1.90	72.93 $\pm$ 0.40	76.09 $\pm$ 0.54	63.47 $\pm$ 0.29
CATD	9.44 $\pm$ 0.00	79.06 $\pm$ 0.48	77.98 $\pm$ 1.37	79.21 $\pm$ 0.46	82.13 $\pm$ 0.23	59.36 $\pm$ 0.07	62.50 $\pm$ 0.56	26.10 $\pm$ 0.00	80.41 $\pm$ 0.05	76.29 $\pm$ 0.36	63.25 $\pm$ 0.23
DS	21.00 $\pm$ 0.00	83.06$\pm$0.00	71.51 $\pm$ 0.00	76.61 $\pm$ 0.00	84.46$\pm$0.00	62.63 $\pm$ 0.00	62.02 $\pm$ 0.00	41.75 $\pm$ 0.00	82.35 $\pm$ 0.00	78.67$\pm$0.00	66.41 $\pm$ 0.00
GLAD	16.90 $\pm$ 0.00	79.47 $\pm$ 0.00	77.98 $\pm$ 0.00	78.81 $\pm$ 0.00	83.45 $\pm$ 0.00	61.04 $\pm$ 0.00	63.43 $\pm$ 0.00	29.57 $\pm$ 0.00	80.30 $\pm$ 0.00	76.57 $\pm$ 0.00	64.75 $\pm$ 0.00
IBCC	10.47 $\pm$ 0.00	73.92 $\pm$ 0.00	71.36 $\pm$ 0.00	76.59 $\pm$ 0.00	83.87 $\pm$ 0.00	63.34 $\pm$ 0.00	65.53 $\pm$ 0.00	21.65 $\pm$ 0.00	69.77 $\pm$ 0.00	77.07 $\pm$ 0.00	61.36 $\pm$ 0.00
CBCC	8.87 $\pm$ 0.00	70.81 $\pm$ 0.00	75.99 $\pm$ 0.00	77.04 $\pm$ 0.00	80.54 $\pm$ 0.00	64.35 $\pm$ 0.00	64.26 $\pm$ 0.00	21.65 $\pm$ 0.00	68.00 $\pm$ 0.00	75.43 $\pm$ 0.00	60.69 $\pm$ 0.00
BWA	15.80 $\pm$ 0.00	81.45 $\pm$ 0.00	<u>79.36$\pm$0.00</u>	78.98 $\pm$ 0.00	83.15 $\pm$ 0.00	59.47 $\pm$ 0.00	59.88 $\pm$ 0.00	28.51 $\pm$ 0.00	82.18 $\pm$ 0.00	76.84 $\pm$ 0.00	64.56 $\pm$ 0.00
TiReMGE	18.77 $\pm$ 0.45	78.82 $\pm$ 0.28	76.20 $\pm$ 1.21	72.22 $\pm$ 0.27	82.47 $\pm$ 0.35	62.43 $\pm$ 0.38	64.32 $\pm$ 0.07	33.99 $\pm$ 1.24	67.54 $\pm$ 0.68	76.85 $\pm$ 0.25	63.36 $\pm$ 0.54
EBCC	10.29 $\pm$ 0.48	79.48 $\pm$ 0.06	71.44 $\pm$ 0.00	79.05 $\pm$ 0.00	<u>84.17$\pm$0.00</u>	61.26 $\pm$ 1.05	62.08 $\pm$ 0.37	22.89 $\pm$ 2.02	74.40 $\pm$ 0.00	78.18 $\pm$ 0.00	62.32 $\pm$ 0.24
GOVERN	9.40 $\pm$ 2.12	80.36 $\pm$ 0.57	78.02 $\pm$ 3.57	78.22 $\pm$ 0.48	82.18 $\pm$ 1.03	65.48$\pm$1.09	43.28 $\pm$ 4.02	25.38 $\pm$ 4.75	89.61$\pm$1.79	76.89 $\pm$ 0.77	62.88 $\pm$ 0.80
PTBCC	26.43 $\pm$ 0.00	80.82 $\pm$ 0.00	79.56$\pm$0.00	<u>79.86$\pm$0.00</u>	82.35 $\pm$ 0.00	<u>64.58$\pm$0.00</u>	<u>68.39$\pm$0.00</u>	55.63$\pm$0.00	81.92 $\pm$ 0.00	<u>78.36$\pm$0.00</u>	69.79 $\pm$ 0.00
CPBCC	36.01$\pm$0.00	<u>81.56$\pm$0.00</u>	79.18 $\pm$ 0.00	80.18$\pm$0.00	83.54 $\pm$ 0.00	64.56 $\pm$ 0.00	68.77$\pm$0.00	<u>42.35$\pm$0.00</u>	<u>84.82$\pm$0.00</u>	77.75 $\pm$ 0.00	69.87$\pm$0.00

Overall, our prototype-driven methods, CPBCC and PTBCC, demonstrate strong superiority over state-of-the-art baselines, with CPBCC and PTBCC boosting the average accuracy from 68.73% to 74.11% and 72.40%, respectively. Besides, the seminal confusion-matrix-based method DS, the representation-learning-based method GOVERN, and BWA also show competitive performance.

5.3. Case Studies on Val7 and MS

Figure 4 and Figure 5 present the prototypes and annotator density inferred by PTBCC and CPBCC for datasets Val7 and MS, respectively.

For MS, we observe that the prototype confusion matrices inferred by PTBCC are of low reliability, with confusion matrices exhibiting weak diagonal dominance. Specifically, both the prototypes inferred by PTBCC and CPBCC demonstrate that annotators show a significant confusion with other classes and class 4, indicating the essential difficulty of this dataset. From Table 2, we find that Val7 is also extremely class-imbalanced, revealing the challenges to learn accurate annotator patterns under imbalanced and difficult tasks.

These phenomena explain the poor performance of existing methods on Val7.

Different from Val7, the inter-prototypes inferred by both PTBCC and CPBCC on the MS dataset are more distinguishable between expert annotators and random annotators, with most annotators demonstrating high classification accuracy under different truths. From Table 2, we find that MS is class-balanced and the data is relatively rich, and most annotators perform better on it.

We demonstrate remarkable performance on Val7 and MS, with CPBCC achieving a 26% accuracy improvement on Val7. Furthermore, CPBCC exhibits more flexibility by allowing annotators more disagreement in patterns under different truths, which is co-achieved by the class-specific prototypes and annotators' corresponding weights. Both PTBCC and CPBCC offer rich insights into the structures of all annotators.

5.4. Ablation Study on CPBCC

When fixing $S = 3$, Table 4 presents the average accuracy of CPBCC for varied d , and the accuracy of Val5 and Val7

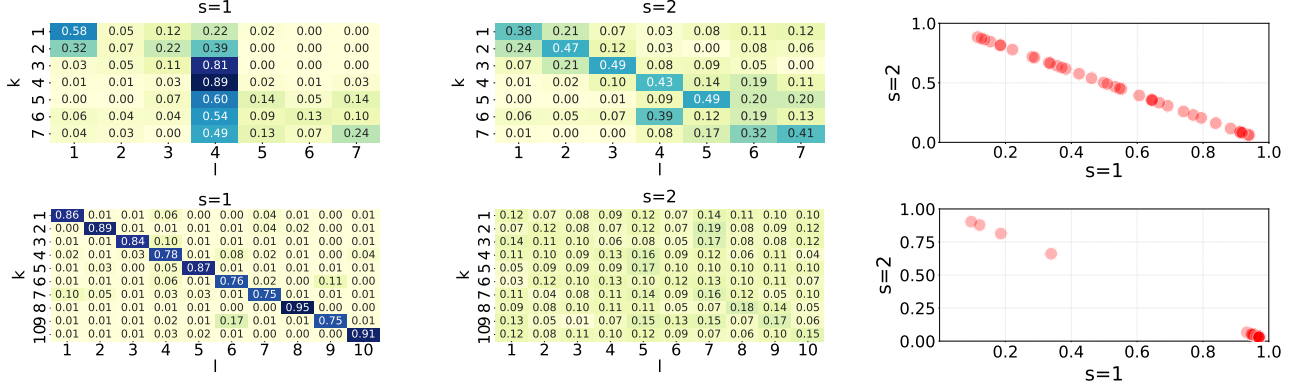


Figure 4. Prototype confusion matrices and annotator density inferred by PTBCC for Val7 and MS, respectively.

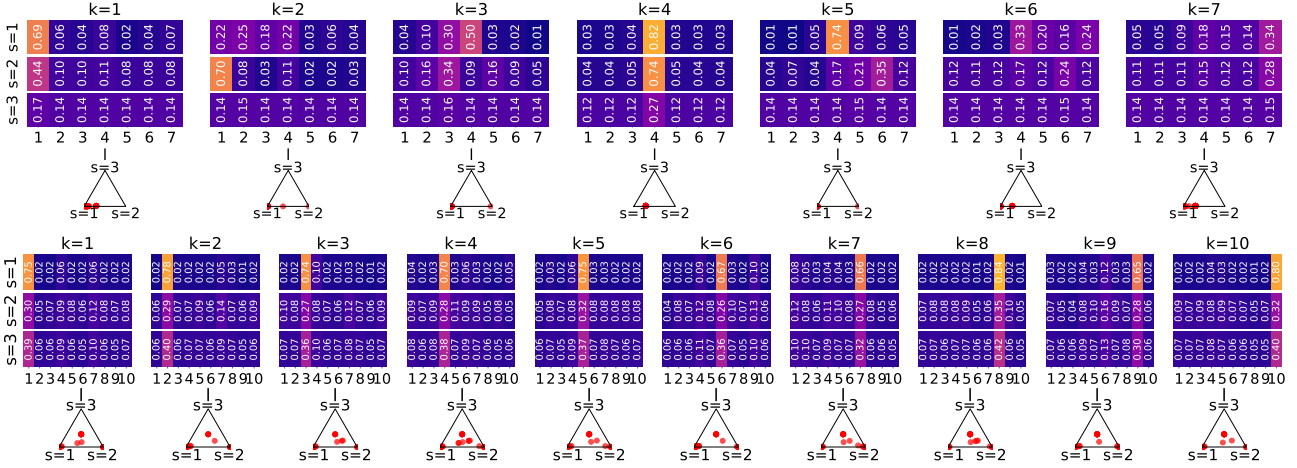


Figure 5. Prototype distributions and annotator density inferred by CPBCC for Val7 and MS, respectively.

for varied b . To investigate the effect of the number of prototypes S , Table 5 presents the average accuracy of CPBCC with parameters initialized using Dirichlet random priors.

Varied d . As Table 4(a) shows, CPBCC exhibits stable performance across a wide range of d . The average accuracy slightly fluctuates as d increases and reaches its peak at 0.35, after which a mild degradation is observed. This indicates that CPBCC is robust to the choice of d , with moderate values yielding optimal performance. This also suggests that setting $\theta_{jik1} = [\phi_{ik}]$ is suitable for most scenarios, except for extremely difficult tasks.

 Table 4. Performance of CPBCC (%) with varied d and b .

 (a) Effect of parameter d on the average accuracy.

d	0.15	0.20	0.25	0.30	0.35	0.40	0.45
Accu	73.80	74.03	73.81	73.66	74.11	73.93	73.85

 (b) Effect of parameter b on datasets Val5 and Val7.

b	3	3.5	4	4.5	5	5.5	6	6.5	7
Val5	48.00	48.00	48.00	48.00	50.00	50.00	47.00	47.00	47.00
Val7	34.00	36.00	36.00	47.00	48.00	48.00	48.00	48.00	48.00

Varied b . Table 4(b) shows the impact of b on Val5 and Val7. Performance on both Val5 and Val7 improves as b increases up to 5.5, with accuracy saturating at $b = 5$ and $b = 5.5$. Beyond this point, accuracy declines as b increases further. These results suggest that moderate diagonal dominance allows CPBCC to better capture complex annotation patterns while maintaining robustness.

Varied S . Table 5 shows the effect of S on average accuracy. As it turns out, increasing the number of prototypes S does not guarantee better performance. Instead, a large S causes data dilution, leading to unstable training and degraded performance. However, despite these challenges, CPBCC demonstrates superior performance compared with the baseline methods.

5.5. Ablation Study on PTBCC

Table 7 presents the performance of PTBCC regarding varied f and m . Notably, we consistently fix $S = 2$ due to the fact that, with more prototypes, the distribution of annotators over prototypes becomes sparser, leading to lower confidence in the prototype confusion matrices. It can be

Table 5. Performance results (mean% $\pm$ std.%) of CPBCC with varied S .

Metric	4	5	6	7	8	9	10
Accuracy (%)	71.51 $\pm$ 0.22	71.27 $\pm$ 0.25	71.36 $\pm$ 0.24	70.95 $\pm$ 0.30	70.77 $\pm$ 0.30	70.66 $\pm$ 0.11	70.24 $\pm$ 0.30
Macro-F1 (%)	67.88 $\pm$ 0.21	67.72 $\pm$ 0.44	67.50 $\pm$ 0.55	67.14 $\pm$ 0.20	67.07 $\pm$ 0.43	66.80 $\pm$ 0.19	66.13 $\pm$ 0.37

Table 6. Performance results (mean% $\pm$ std.%) of PTBCC with varied S .

Metric	3	4	5	6	7	8
Accuracy	69.94 $\pm$ 0.75	69.44 $\pm$ 0.68	69.03 $\pm$ 0.49	68.78 $\pm$ 0.28	69.01 $\pm$ 0.45	68.95 $\pm$ 0.39
Macro-F1	67.07 $\pm$ 0.79	65.74 $\pm$ 0.36	65.53 $\pm$ 0.34	65.32 $\pm$ 0.33	65.66 $\pm$ 0.16	65.46 $\pm$ 0.35

observed that under varied f or m , PTBCC consistently outperforms state-of-the-art methods, demonstrating the stability of training two prototypes. Moreover, the results indicate that we should offer initial prototypes with abundant differentiability to yield good training performance.

To investigate the effect of the number of prototypes S , Table 6 presents the average accuracy of PTBCC with parameters initialized using Dirichlet random priors. As observed in CPTBCC, increasing the number of prototypes does not lead to performance improvements.

Table 7. Average accuracy (%) of PTBCC with varied f and m .

f	2	3	4	5	6	7	8
Accu	71.12	72.08	72.12	72.40	72.34	72.28	72.32
m	1.1	1.15	1.2	1.25	1.3	1.35	1.4
Accu	72.10	72.03	72.11	72.13	72.34	72.40	72.27

6. Efficiency

Figure 6 presents the runtime on 10 datasets in ascending order of the annotation size. The runtime of all methods generally increases with the number of annotations. MV, as the most naive method, consistently ranks first. BWA, as the best-performed single-parameter-based method, follows MV. Our CPBCC and PTBCC also rank in the upper tier, with PTBCC ranking third most of the time and maintaining a clear superiority over confusion-matrix-based methods other than DS.

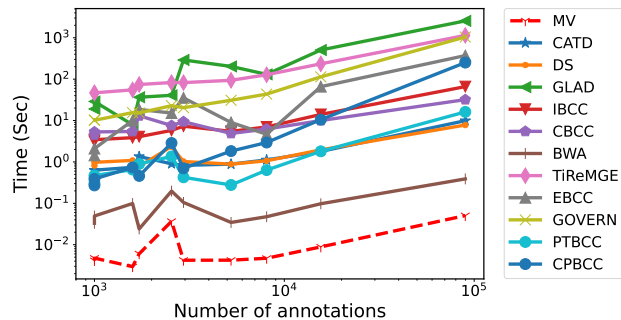


Figure 6. Comparison of running time on 10 datasets.

DS updates via maximum likelihood estimation, making it more efficient than EBCC and FGBCC, which involve fine-grained inference. GLAD exhibits the lowest efficiency; we attribute this to its incorporation of task difficulty and the use of gradient-based inference. TiReMGE demands a long training time, primarily because of the optimization of an objective function capturing complex annotator-task relationships.

Overall, our CPBCC and PTBCC exhibit competitive efficiency, with the performance of PTBCC demonstrating outstanding superiority in most cases. Moreover, confusion-matrix-based and representation-learning-based methods tend to require longer training time.

7. Conclusion

This paper addresses the data sparsity, class imbalance, and insufficient expressiveness problems that fundamentally bottleneck SOTA truth inference. In response, we propose CPBCC (Class-specific Prototype-driven Bayesian Classifier Combination). Unlike existing methods that train a separate confusion matrix for each annotator, CPBCC creatively captures annotators by: (i) learning class-specific prototype annotation patterns across all annotators, and (ii) learning annotator-specific weights over prototypes. This framework systematically addresses the issues of data sparsity and class imbalance, while extending annotator characterization from an individual confusion matrix per annotator to a personalized distribution over diverse prototypes. Extensive experiments across 10 real-world datasets exhibiting diverse data characteristics demonstrate that we achieve a 26% accuracy improvement in the best case, and a 5.38% increase in average accuracy. For future work, we wish to explore the multi-model annotation aggregation.

Acknowledgements

This work was supported by National Key R&D Program of China (Grant Number 2023YFC3209203).

Impact Statement

This paper aims to advance machine learning and fact-checking in the era of large language models by improving the accuracy of crowdsourcing annotations. To the best of our knowledge, it raises no ethical concerns.

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A. Detailed related work

Depending on how annotators are modeled, existing works can be categorized into single-parameter-based, confusion-matrix-based, and representation-learning-based approaches.

Single-parameter-based Methods assume each annotator keeps a consistent reliability across tasks with different truths. Whitehill et al. (2009) model truth log-odds via a bilinear form of task difficulty and annotator reliability. Li et al. (2014b) model annotator errors using the Gaussian distribution and exploit the upper bound of the confidence interval to derive annotator reliability. (Aydin et al., 2014) also minimizes a weighted sum of annotators’ errors, constraining annotator reliability to be non-negative and summing to one, and infers using block coordinate descent techniques. Donald & Combes (2017) derives a lower bound on the minimax estimation error and proposes triangular estimation for annotator reliability, which provides non-asymptotic performance guarantees. Li et al. (2019b) and Cao et al. (2020) model annotator reliability using Gamma and scaled inverse chi-squared distribution, and updating them via the expectation-maximization and gradient descent method, respectively. Most single-parameter-based methods are computationally efficient but less robust due to their strong homogeneity assumption.

Confusion-matrix-based Methods learn an individual confusion matrix per annotator to capture class-specific annotator reliability. This idea originates from the DS model (Dawid & Skene, 1979), which is the first to capture annotator reliability by confusion matrices. Kim & Ghahramani (2012) extend DS (Dawid & Skene, 1979) by incorporating Dirichlet priors. Venanzi et al. (2014) propose detecting communities among annotators to make annotators within the same community similar. Li et al. (2019a) classify tasks with the same truth into subtypes to capture annotator correlation. Ibrahim et al. (2019); Ibrahim & Fu (2021) exploit matrix factorization and annotators’ co-occurrences to infer confusion matrices and truths alternately. Chen et al. (2025) develop a Softmax-Gaussian structure to replace the Dirichlet-Multinomial pattern of (Kim & Ghahramani, 2012). Song et al. (2021) propose detecting the copying behaviors between annotators to penalize the copiers. Confusion-matrix-based methods generally perform better and demonstrate greater potential. However, they are sensitive to data sparsity and class imbalance, and are still limited in capturing annotators’ full reliability patterns.

Representation-learning-based Methods leverage representation learning to encode relations among annotators, tasks, and annotations. Yin et al. (2017) propose LAA, which integrates a classifier-encoder and a reconstructor-decoder, and treats annotations as latent features. Wu et al. (2023a) introduce a multi-view graph embedding model over the task-annotator graph, and propose a heritable mechanism for annotator estimation. Wu et al. (2023b) proposes a hierarchical graph autoencoder to capture complex crowdsourcing relations on a task-annotator graph from diverse views, and develops a vector initialization method guided by annotator reliability for the representation of tasks and annotators. Liu et al. (2024) propose a graph contrastive framework, where data augmentation and a contrastive objective are introduced to encourage invariant representations across views. Despite their expressiveness, representation-learning-based methods require abundant data for network training, which constrains their robustness in sparse settings.

Besides, it should also be noted that there is a wide spectrum of related works focusing on settings where rich task features are available (Chen et al., 2022b; Jiang et al., 2022), and studies focusing on mobile crowdsourcing where annotators wear mobile devices to perform sensing tasks within specific conditions (Xu et al., 2023; Cheng et al., 2023), and learning from crowds with noisy annotations where the final goal is to training a classifier (Chu et al., 2021b; Cao et al., 2023).

B. The inference details of PTBCC

The joint distribution of PTBCC is:

$$\begin{aligned} p(Y, Z, V, X, \pi, \tau | u, \beta, a) &= p(\tau | u) p(\pi | \beta) p(V | a) p(Z | \tau) p(X | \pi) p(Y | X, Z, V) \\ &= \text{Dir}(\vec{\tau} | \vec{u}) \cdot \prod_j \text{Dir}(\vec{\pi}_j | \vec{\beta}_j) \cdot \prod_s \prod_k \text{Dir}(\vec{v}_{sk} | \vec{a}_{sk}) \cdot \prod_i \tau_{z_i} \prod_{j \in \mathcal{W}_i} \pi_{j x_{ji}} v_{x_{ji} z_i} y_{ij} \end{aligned} \quad (17)$$

Since direct inference is computationally intractable, we employ the mean-field approximation in variational inference (Bishop, 2006) to approximate the posterior. Therefore, the goal is to find a variational distribution q ,

$$q(Z, V, X, \pi, \tau) = \text{Dir}(\vec{\tau} | \vec{v}) \cdot \prod_j \text{Dir}(\vec{\pi}_j | \vec{\eta}_j) \cdot \prod_s \prod_k \text{Dir}(\vec{v}_{sk} | \vec{\mu}_{sk}) \cdot \prod_i q(z_i | \vec{\phi}_i) \prod_{j \in \mathcal{W}_i} q(x_{ji} | \vec{\theta}_{ji}) \quad (18)$$

To this end, the lower bound of the log evidence is:

$$\begin{aligned}
 \log L(q) = & \sum_k (u_k - \nu_k + \sum_i \phi_{ik}) \mathbb{E}_q \log \tau_k + \log B(\vec{\nu}) + \sum_j \{ \sum_s (\beta_{js} - \eta_{js} + \sum_{i \in \mathcal{T}_j} \theta_{jis}) \mathbb{E}_q \log \pi_{js} + \log B(\vec{\eta}_j) \} \\
 & + \sum_s \sum_k \{ \sum_l (a_{skl} - \mu_{skl} + \sum_j \sum_{i \in \mathcal{T}_j} \phi_{ik} \theta_{jis} \mathbf{1}[y_{ij} = l]) \mathbb{E}_q \log v_{skl} + \log B(\vec{\mu}_{sk}) \} \\
 & - \sum_i \sum_k \phi_{ik} \log \phi_{ik} - \sum_j \sum_{i \in \mathcal{T}_j} \sum_s \theta_{jis} \log \theta_{jis} + \text{Const}
 \end{aligned} \tag{19}$$

We sequentially derive each term in Equation (18) by maximizing Equation (19). Following the standard mean-field update steps, the update rules are:

$$\nu_k = u_k + \sum_i \phi_{ik} \tag{20}$$

$$\eta_{js} = \beta_{js} + \sum_{i \in \mathcal{T}_j} \theta_{jis} \tag{21}$$

$$\mu_{skl} = a_{skl} + \sum_j \sum_{i \in \mathcal{T}_j} \phi_{ik} \theta_{jis} \mathbf{1}[y_{ij} = l] \tag{22}$$

$$\theta_{jis} \propto \exp(\mathbb{E}_q \log \pi_{js} + \sum_k \phi_{ik} \mathbb{E}_q \log v_{sk y_{ij}}) \tag{23}$$

$$\phi_{ik} \propto \exp(\mathbb{E}_q \log \tau_k + \sum_{j \in \mathcal{W}_i} \sum_s \theta_{jis} \mathbb{E}_q \log v_{sk y_{ij}}) \tag{24}$$

and the expectations of the log parameters are:

$$\mathbb{E}_q \log \tau_k = \psi(\nu_k) - \psi(\sum_k \nu_k) \tag{25}$$

$$\mathbb{E}_q \log \pi_{js} = \psi(\eta_{js}) - \psi(\sum_s \eta_{js}) \tag{26}$$

$$\mathbb{E}_q \log v_{skl} = \psi(\mu_{skl}) - \psi(\sum_l \mu_{skl}) \tag{27}$$

where $\psi(\cdot)$ denotes digamma function. We update all parameters iteratively until the change in each ϕ_{ik} falls below a threshold ξ .

C. Proof of Theorem 1

Proof. Our proof builds upon the minimax error rate bound developed by Imamura et al. (2018) for DS (Dawid & Skene, 1979). Specifically, the minimax error rate of DS satisfies

$$\inf_Z \sup \mathbb{E}[\mathcal{L}(\hat{Z}, Z)] \geq -\frac{\log 2}{T^2 \log K} + \frac{1}{T^2 \log K} \sum_i \left(H(\phi_i) - \sum_{j \in \mathcal{W}_i} \sum_{k, k'} \phi_{ik} \phi_{ik'} \text{KL}(\vec{\Psi}_{jk} || \vec{\Psi}_{jk'}) \right) \tag{28}$$

where $H(\vec{\phi}_i) = \sum_k \phi_{ik} \log \phi_{ik}$ and $\vec{\Psi}_{jk}$ ($k \in \{1 \dots K\}$) denotes the confusion matrix of w_j . We now specialize this bound to PTBCC and CPBCC by characterizing $\vec{\Phi}_{jk}$ under each model and upper bounding the row-wise Kullback–Leibler divergence.

In PTBCC, the effective confusion matrix satisfies

$$\vec{\Psi}_{jk} = \sum_s \pi_{js} \vec{v}_{sk}. \tag{29}$$

For any pair of classes k and k' , we have

$$\begin{aligned} \text{KL}(\vec{\Psi}_{jk} || \vec{\Psi}_{jk'}) &= \text{KL}\left(\sum_s \pi_{js} \vec{v}_{sk} || \sum_s \pi_{js} \vec{v}_{sk'}\right) \\ &\leq \sum_s \pi_{js} \text{KL}(\vec{v}_{sk} || \vec{v}_{sk'}) \end{aligned} \quad (30)$$

where the inequality is derived by the log-sum inequality. Substituting (30) into (28) yields equation (15).

In CPBCC, the prototype distribution depends on the true class. Therefore, the effective confusion matrix satisfies

$$\vec{\Psi}_{jk} = \sum_s \pi_{jks} \vec{v}_{ks} \quad (31)$$

For any k and k' , applying the log-sum inequality gives

$$\begin{aligned} \text{KL}(\vec{\Psi}_{jk} || \vec{\Psi}_{jk'}) &= \text{KL}\left(\sum_s \pi_{jks} \vec{v}_{ks} || \sum_s \pi_{jk's} \vec{v}_{k's}\right) \\ &\leq \sum_s \pi_{jks} \text{KL}(\vec{v}_{ks} || \vec{v}_{k's}) \end{aligned} \quad (32)$$

Substituting (32) into (28) yields equation (16).

Therefore, the theorem gets proved. $\square$

D. Limitation

Because this paper targets data-sparse and class-imbalanced crowdsourcing settings that are prevalent in real-world settings, it may lag behind representation-learning-based methods in data-abundant and class-balanced scenarios.